Arihan Yadav

Computer Engineering and Computer Science
Cupertino, CA | (408) 892 7275 | apyadav@wisc.edu | https://www.linkedin.com/in/arihanyadav | https://air-crypto.github.io | US Citizen

EDUCATION

University of Wisconsin-Madison, December 2026
Bachelor of Science, GPA: 3.5
Double Major in Computer Engineering & Computer Science

EXPERIENCE

AMD (Advanced Micro Devices), Boxborough, MA - AI/ML Intern
Post-Training LLMs with Reinforcement Learning for Proprietary Code Generation
May 2026 - August 2026

- Built an end-to-end CPT, SFT, RL post-training pipeline to specialize a base model for proprietary device code generation: continued pretraining on the target codebase, supervised fine-tuning on reasoning traces and tool-use trajectories distilled from a larger teacher model, then reinforcement learning with GSPO and verifiable rewards (RLVR).
- Trained a model that passed more directed device tests than GPT and Claude Opus while generating solutions in under 12k tokens, versus 102k for Opus and 360k for GPT, an **8x to 30x reduction in token cost at higher accuracy.**
- Engineered a fused cross-entropy kernel with near-constant memory as sequence length grows, unlocking an 8x larger training context window than the prior implementation allowed.
- Architected distributed training and inference across 16 MI250 GPUs (two 8-GPU clusters over RCCL), separating a dedicated training server from a vLLM inference server with speculative decoding and synchronizing policy weights every 3 steps.
- Measured reward hacking and grader-process exploitation in the RL loop and implemented safeguards to harden the reward and grading infrastructure against both.
- **First author of 2 papers published at AMD's Global Technical Authors Conference** detailing the RL methodology, and building harnesses to train and evaluate models.

AMD (Advanced Micro Devices), Boxborough, MA - AI/ML Intern
Developing AI Frameworks and Models to Accelerate Hardware Design Efficiency
May 2025 - August 2025

- Developed an AI pipeline for root-cause analysis of hardware traces. Clustered failures by root causes rather than just error signatures with 82% accuracy (vs 28% for state of the art systems) against internal benchmarks. Reduced verification time and resources by 30x in production.
- Built a data pipeline from a Dremio data lake to a GPU cluster. Processed 97k transactions/second on 3.5B-trace corpus at almost linear scale.
- Redesigned LLM architecture from hierarchical full attention to sliding-window attention, boosting memory efficiency (exponential to linear) for long sequences while preserving causality and recent-event focus.
- Ran ablation tests to configure QLoRA quantization for 4-bit training, achieving 22x context window growth (from 512 to 11,000 tokens) resulting in a 54% accuracy boost.
- Implemented fine-tuning with layer freezing, enabling quick adaptation to new CPU/GPU chip families with minimal new training data while maintaining low reconstruction loss and sharp cluster separation.
- Ported the end-to-end pipeline from Python/PyTorch to C++ (LibTorch/pybind11), fusing hot paths and optimizing memory layout/data movement to deliver 4.6x faster training and 3x faster inference with custom HIP kernels.
- **First author of a paper published at AMD's Global Technical Authors Conference**, highlighting superior results from quantized models leveraging sliding window attention.

Applied Research Laboratory for Intelligence and Security (ARLIS), UW Madison, WI & UMD, MD **-** *Researcher (Remote)*
Building a Multimodal Search Engine using LLMs and RAG for the Department of Defense (DoD) unifying information retrieval across diverse input data types (text, images, audio, and video)
November 2024 - Present

- Lead developer working with the DoD via the ARLIS institute on novel context-aware multi-modal retrieval systems integrating LLMs with Retrieval Augmented Generation(RAG). This search engine is used for naval assets, declassification workflows, and other defense use cases.
- Architected and trained a PyTorch-based model that aligns heterogeneous data embeddings into a unified semantic space, enabling cross-modal retrieval.
- Achieved an accuracy of 94.53%, Top-5 accuracy of 99.93%, and Mean Reciprocal Rank (MRR) of 0.9703, outperforming other state-of-the-art models such as CLIP (35.78% accuracy) on image-caption retrieval.

Molecular Imaging/Magnetic Resonance Technology Lab, Madison, WI **-** *Undergraduate Researcher*
Driving AI Innovations in Medicine Through Novel Work in Multimodal LLMs enhancing diagnostic accuracy and decision-making, critical for accurate patient diagnoses.

January 2024 - Present

- Previously, worked on a custom multi-modal Projection Model (Pytorch) to translate pseudocode embeddings to C code embeddings. Improved retrieval accuracy by 58.2% compared to state-of-the-art models and algorithms (e.g., BM25 and Sentence Transformers). **Published a paper as the primary author. Additionally, UW-Madison is in the process of filing a patent for this innovation.**
- Formerly partnered with a postdoc to address memory limitations encountered during fine-tuning LLMs like LLaMA, significantly reducing the memory footprint of training pipelines.
    - Achieved a 41.6% reduction in memory footprint (from 212 GB to 124 GB) for LLM training across 12 A100 NVIDIA GPUs (leveraging HTCondor), enabling larger models and batch sizes.

Theom Inc., San Jose, CA - *Software Developer Intern*

Built High-Performance Fine-Grained Access Control for LLMs with Vector Search. Plus, a novel Data lineage Tracker at Scale

June 2024 - December 2024

- Designed and implemented a Flask-based RAG service coupled with an LLM, integrating multiple vector databases (Pinecone, OpenSearch, Milvus, Chroma, and MongoDB) to enable efficient similarity searches based on the identity & context of who is querying.
    - Engineered novel algorithms to create, update, and delete vector embeddings across multiple databases, supporting batch processing for large-scale data ingestion from source data systems with appropriate permissions per embedding node in the vector.
    - Created a mechanism to load and manage equivalent permission classifications, allowing the vector system to filter results based on the querier's access policies, context, and identity.
    - Utilized LangChain and LlamaIndex to scale deployment of the system to customers, by removing the need for individual custom frameworks in favor of generalized scalable solutions.
- Worked in a fast-paced startup environment and was given the objective of improving runtime performance of data classification by 10x and tracking data lineage (traceability) as data gets modified, moved around, and replicated inside the company's data stores.
    - Implemented a semantic search algorithm using C++, SQL, and AWS OpenSearch that identified similar data, significantly outperforming state-of-the-art uses of fuzzy hashing.
    - Met the runtime performance objective and improved data traceability by 572%.

Wisconsin Robotics, Madison, WI - *Team Member*

Enhancing Autonomous Vehicle Navigation through Sensor Fusion

September 2023 - Present

- Reduced navigation errors by 34% and enhanced path-planning efficiency by 26% of an autonomous vehicle running a real-time operating system by developing new sensor-fusion (LiDAR and camera) based trajectory models.
- Leveraging tools like Docker and GitHub alongside ROS to containerize applications and facilitate seamless communication between software systems and robots.

PROJECTS

VLIW SIMD Processor Optimization, Madison, WI - Researcher and Developer

February 2026 - February 2026

- Optimized a VLIW SIMD processor simulator kernel to 1336 cycles, surpassing Claude Opus 4.5's best performance (1363 cycles) and achieving a 110.6x speedup over the 147,734-cycle baseline through aggressive optimization.
- Implemented selective tree node preloading to eliminate 87% of memory loads via nested vselect multiplexers and custom dependency-aware VLIW instruction scheduler maximizing parallel execution. This way we detect hazards and dependencies quickly and correctly.

FPGA-Based eBike Controller, Madison, WI - Controller *Team Lead*

February 2025 - May 2025

- Developed a high performance, battery efficient, digital eBike controller using System Verilog, integrating PID-based motor drive logic, SPI communication interfaces, exponential averaging filters, inertial sensor fusion, and UART telemetry for real-time sensor data transmission.
- Integrated pipeline stages and synthesis optimizations to meet a 400 MHz clock requirement, while achieving a 0.3ns setup slack margin.
- Reduced synthesized hardware area from 20k μm² to 13k μm² (Synopsys 32nm library) through logic and synthesis script optimizations.
- Designed testbenches using constrained random verification, and compiled it onto an FPGA mounted on a bike to demonstrate.

ACHIEVEMENTS

USA Computing Olympiad - Gold Division

Patent granted for high accuracy, low latency American Sign Language to Text translator that runs on mobile devices.

Selected Coursework: AI-Assisted Software Dev, World Models, System Architecture for Quantum Computers, Matrix Methods in Machine Learning, Artificial Neural Networks, Operating Systems, System Synthesis and Design, Data Structures and Algorithms, Database Design and Management, Machine Organization and Programming, Random Signal Analysis and Statistics, Signals, Information, and Computation

Programming Languages: Java, C, C++, Python, Go, JavaScript, MATLAB, SQL, Assembly, Verilog

Frameworks: PyTorch, JAX, TensorFlow, LangChain, LlamaIndex, Flask, AWS OpenSearch, GitHub, ROCm, CUDA, Docker, ROS, MLIR